Review addendum: PR #1168

Date: 2026-10-29. Add fast local polynomial-gcd gate for C–E fibers.

Original author head: 0e53ef8dd1b290f1cdba82d530d8f4f2e2f0e192. The original study and every frozen source, protocol, result, failed attempt and timing cap are preserved. This addendum takes precedence over unqualified experimental-status language in the archived report.

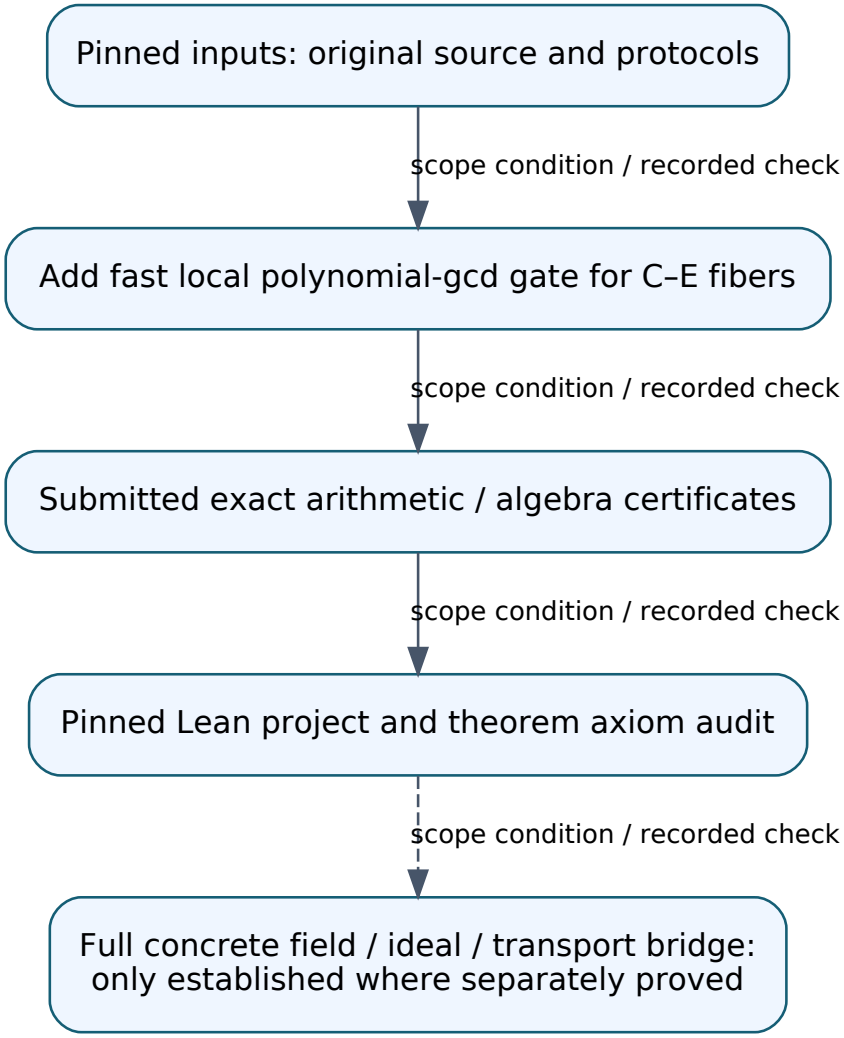
Recorded finding and its scope

A finite-field polynomial gcd gate reproduces the exact C–E pair counts at **all eleven** global-norm exceptions below ten million. It processes the 22 rational source-j=1728 C points with **501 polynomial multiplications and 811 polynomial remainders** in total; the largest individual E-fiber uses 48 polynomial multiplications. These are counted operations in the new fixed-degree E-fiber kernel. The prior direct checks of small exceptional cells scanned field values, while this kernel uses modular polynomial exponentiation whose multiplication count grows with log p for fixed resolvent degree. Source-point extraction is a separate degree-18 finite-field factorization step and is excluded from those kernel counts.

The paragraph above describes the retained certificate record, not a new experiment. Source: experiments/hecke-432-j1728-fast-local-root-gate-20261026/RESULT.md (../../experiments/hecke-432-j1728-fast-local-root-gate-20261026/RESULT.md). This review treats publication of protocol and results together as retrospective/exploratory unless an independent earlier publication witness is supplied. An archived PASS_PROSPECTIVE status is a literal historical label, not evidence of pre-observation registration. In particular, PR #1164's six-row holdout and PR #1190's 62/62 correctness checks retain their numerical results with this qualification.

Evidence-linked explanation

Evidence and proof boundaries for this study. Symbols C, B, E and K denote the frozen coordinate models or coefficient field, not newly registered subgroup instances. This diagram records algebra/certificate dependencies; it does not introduce a verified isogeny edge, EC1 identity, ISO1 route, or ECDLP transport claim.



Editable source: 1168.dot (1168.dot). Report and visual: 1168.pdf (1168.pdf).

Replay and proof map

The exact manifest binds 13 files in this study. The shared project pins Lean leanprover/lean4:v4.35.0-rc4 and Mathlib 021c46bdd042f67b538b9f9871d556a93713a9a9. All 37 original modules, 113 requested declaration reports and 142 noninternal theorems passed the integrated proof replay; this does not formalize every concrete Sage/PARI field or ideal interpretation. The source manifest and proof map are tools/ci/quartic_certificates.json (../../tools/ci/quartic_certificates.json) and the original integration report (../ci-repair-quartic-20261027.md).

This study's original modules: experiments/hecke-432-j1728-fast-local-root-gate-20261026/formal/FrobeniusGcdGate.lean.

A new portable replay ran all 42 original integer auditors successfully in a disposable canonical-layout container. The checkout was read-only and all 1,404 original input hashes were verified before and after replay. See portable replay evidence (portable-replay/results.json). The separate field-index successor runner uses new output directories, unchanged source/protocol bytes, original timeout limits and the checked Sage launcher. Preparation and I/O controls are not a new Sage experiment.

Remaining obligations and graph review

Arithmetic table certificates and universal ring/permutation identities retain their precise hypotheses. Concrete number-field encodings, integral bases, ideals, Hensel lifting and Frobenius interpretation require their own bridges. No security, speedup, prospective-validation or general impossibility claim is added.

The candidate catalog, volcano-IC figure set and paper figures were checked for affected links. This is a review and portability correction with no new surviving graphable mathematical fact; canonical graphs remain unchanged. The precise checked-path inventory is graph-review.json (graph-review.json).